

Dioxins (PCDD/Fs) and PCBs in offal: occurrence and dietary exposure.

Offals are widely consumed in different cuisines, but information on the occurrence of dibenzo-p-dioxins, dibenzofurans (PCDD/Fs) and polychlorinated biphenyls (PCBs) in these foods is sparse. In the first structured investigation of its kind, this study reports levels of these contaminants in commonly consumed offals (n=173) such as lamb, ox, deer and pig's liver, kidneys, tongue and heart, and offal products such as pâté, haggis, tripe and black pudding. The results support literature observations on the preferential accumulation of contaminants in liver tissue, as the highest concentrations of PCDD/Fs were observed in liver, relative to the other organs (e.g. 8.4 ng WHO-TEQ kg⁽⁻¹⁾ lamb liver compared to 1.1 ng WHO-TEQ kg⁽⁻¹⁾ lamb kidney and 1.27 ng WHO-TEQ kg⁽⁻¹⁾ lamb heart). Offal products generally showed lower contaminant levels which may be a result of processing or dilution. For most samples, the main contribution to WHO-TEQ arose from PCDD/Fs rather than PCBs. Just under half of the lamb liver samples showed PCDD/F concentrations that exceeded the EU maximum limit of 6 ng kg⁽⁻¹⁾ fat weight (although deer liver which is not subject to the regulation, generally showed higher levels). Dietary exposure estimates indicate that the weekly consumption of up to two 100g portions of lamb, ox, calf or pig liver or one portion of deer liver would not breach the tolerable daily intake (TDI) level even when the rest of the diet was included. However, the consumption of more than one portion of deer liver per week may lead to the TDI being exceeded. Crown Copyright © 2010. Published by Elsevier Ltd. All rights reserved.